

202512_morph_lm

```
## Loading required package: RRPP

## Loading required package: rgl

## Warning in rgl.init(initValue, onlyNULL): RGL: unable to open X11 display

## Warning: 'rgl.init' failed, will use the null device.
## See '?rgl.useNULL' for ways to avoid this warning.

## Loading required package: Matrix

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1      v readr 2.1.5
## v ggplot2 4.0.0      v stringr 1.5.2
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.1.0       v tidyr 1.3.1
## -- Conflicts ----- tidyverse_conflicts() --
## x tidyr::expand() masks Matrix::expand()
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## x tidyr::pack()   masks Matrix::pack()
## x tidyr::unpack() masks Matrix::unpack()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'gridExtra'
##
##
## The following object is masked from 'package:dplyr':
##
##   combine
##
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
##
## Loading required package: mclust
##
## Package 'mclust' version 6.1.1
## Type 'citation("mclust")' for citing this R package in publications.
##
##
## Attaching package: 'mclust'
##
##
## The following object is masked from 'package:purrr':
##
##     map
##
## Package 'clustvarsel' version 2.3.5
##
## Type 'citation("clustvarsel")' for citing this R package in publications.
```

Import data file

Dir:“/Volumes/Yinghui/202512-dev-seq-analyses”

Plot setting

```
#linear measurements
```

Removing the value of lobate and auricle when those structures are not present standard: ALO 0.07mm AW 0.07mm LTM 0.07mm LL 0.07mm

convert the sample_id to date

change the sample id to individual

allometric dev visualisation

obtain the natural log of the value

Common practice when studying allometric development because it is easier to deal with proportional variation.

plot allometric dev

Using body length as the x axis only focusing on the 20C group

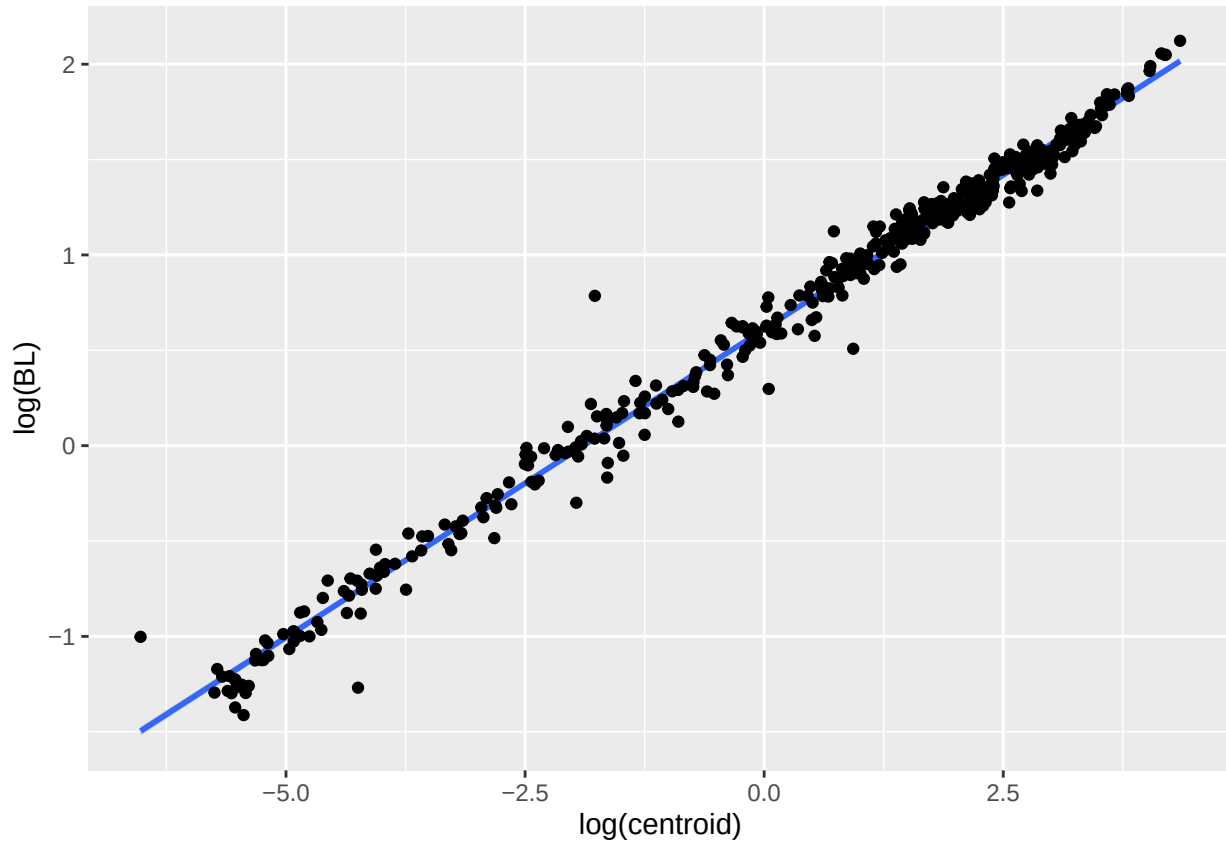
plot by individual

plot all measurements together on centroid

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

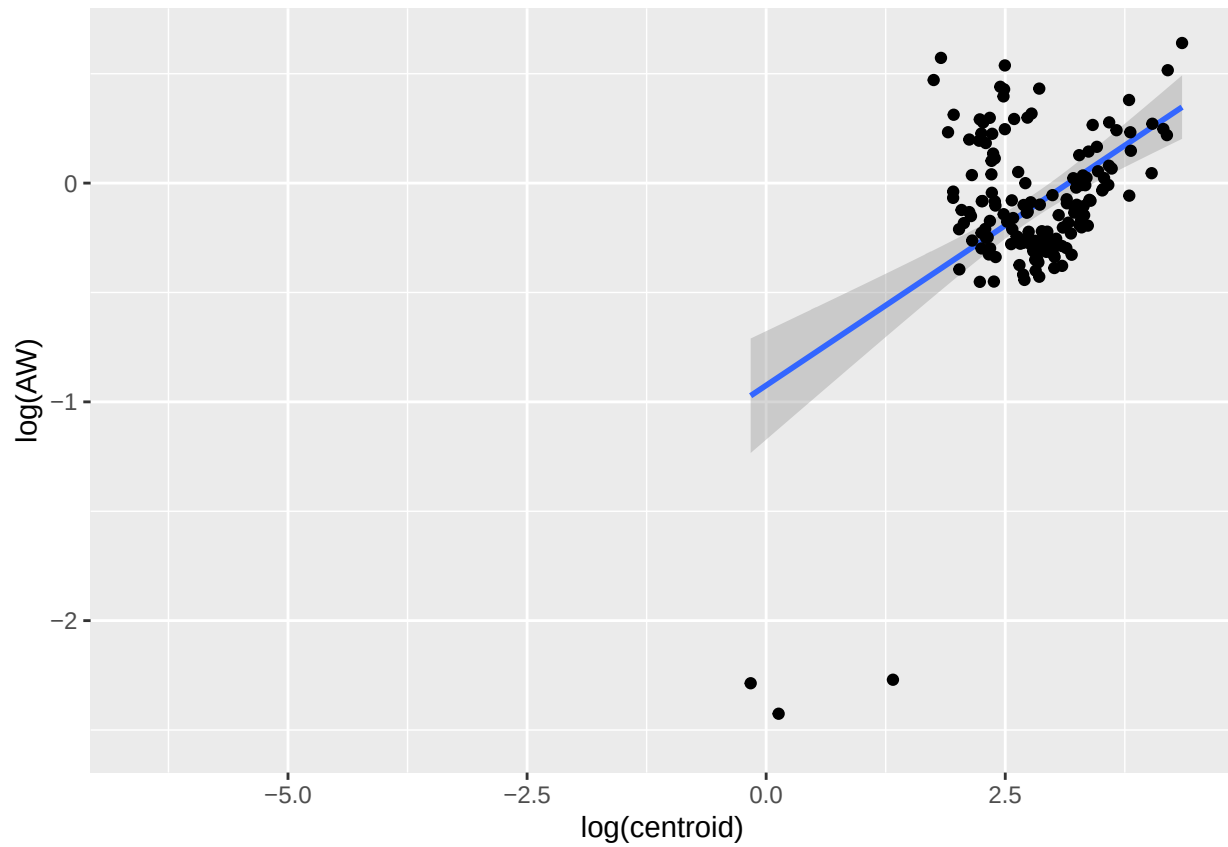
```
## Warning: Removed 9 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 297 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

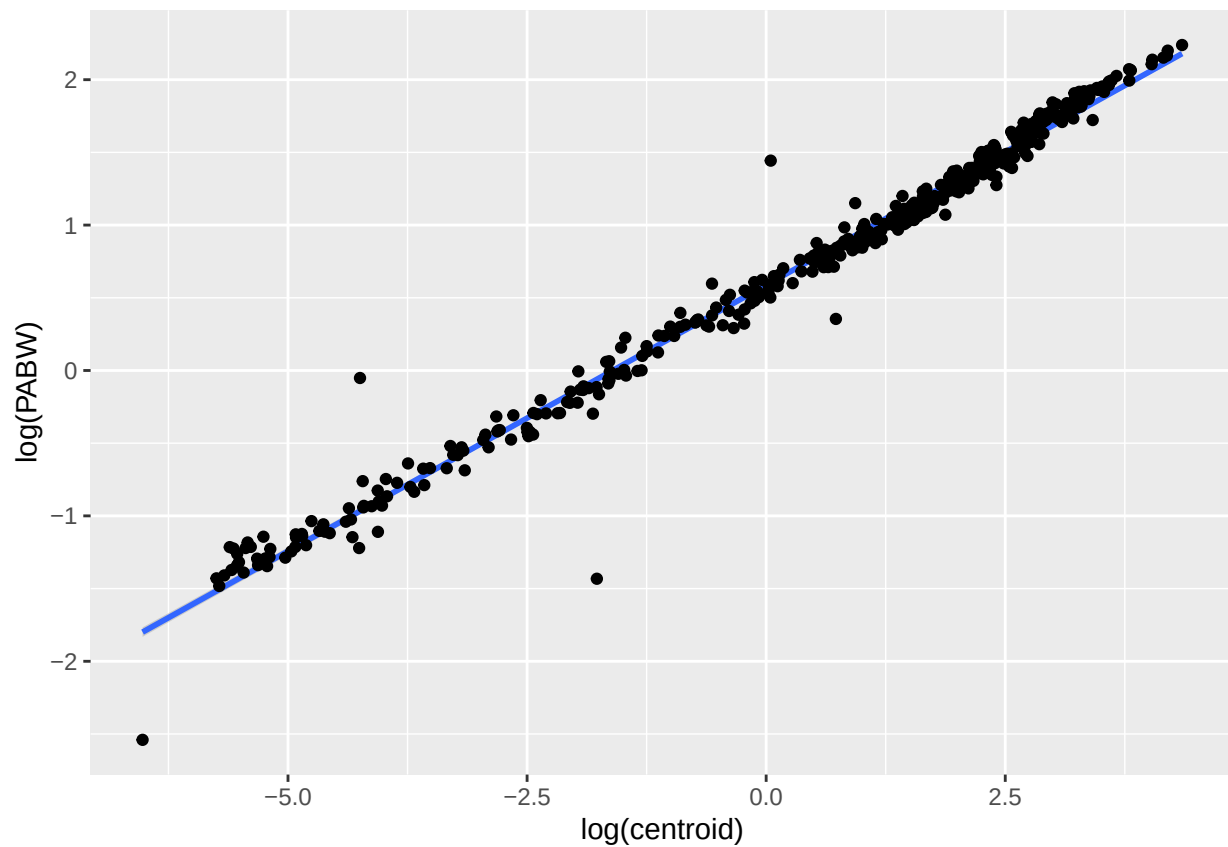
```
## Warning: Removed 297 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

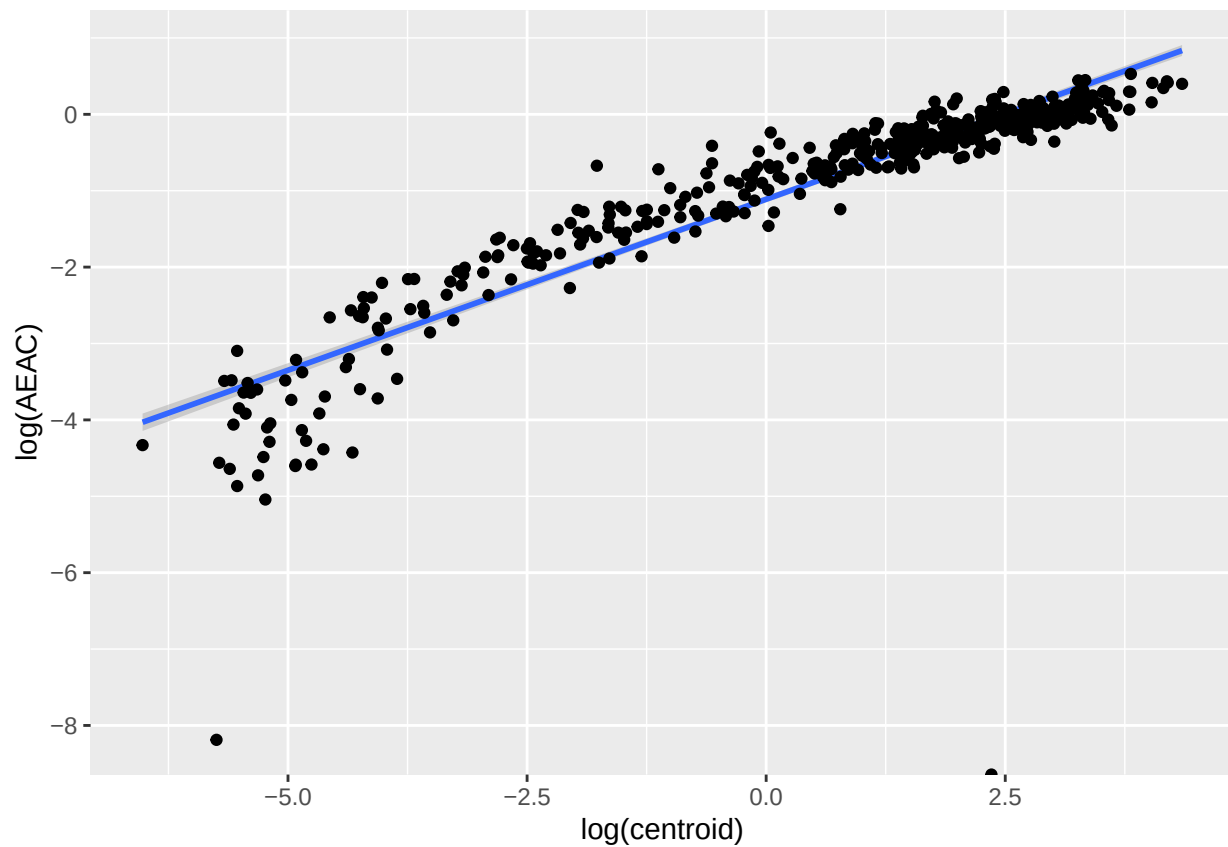
```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 19 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

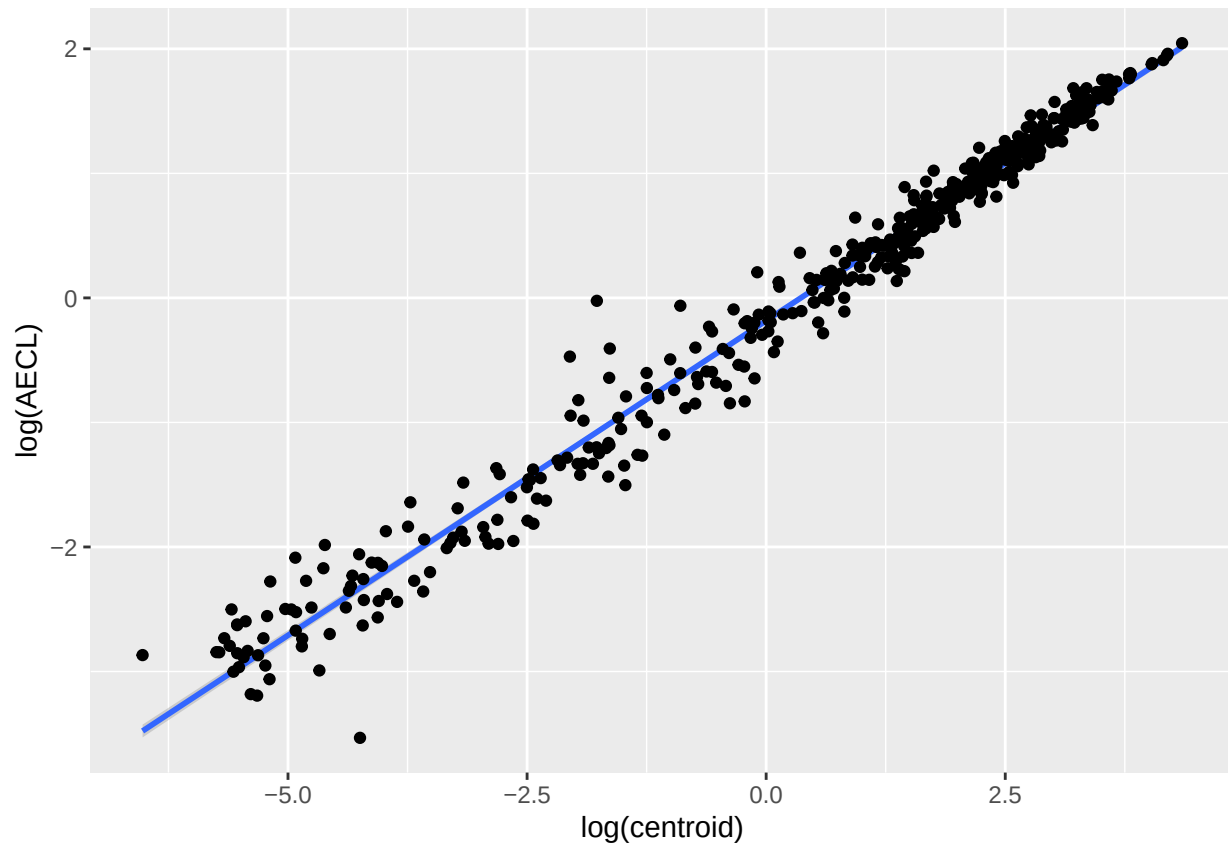
```
## Warning: Removed 18 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

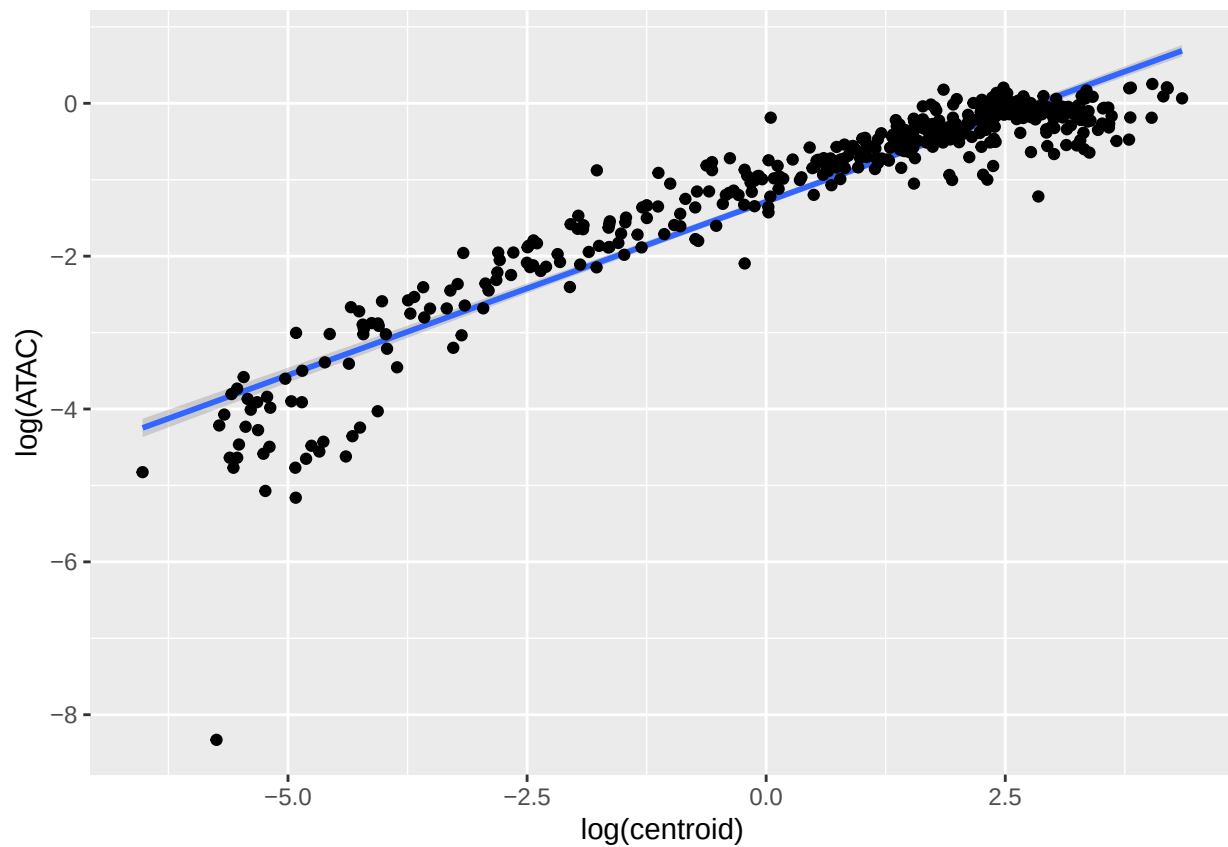
```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 18 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

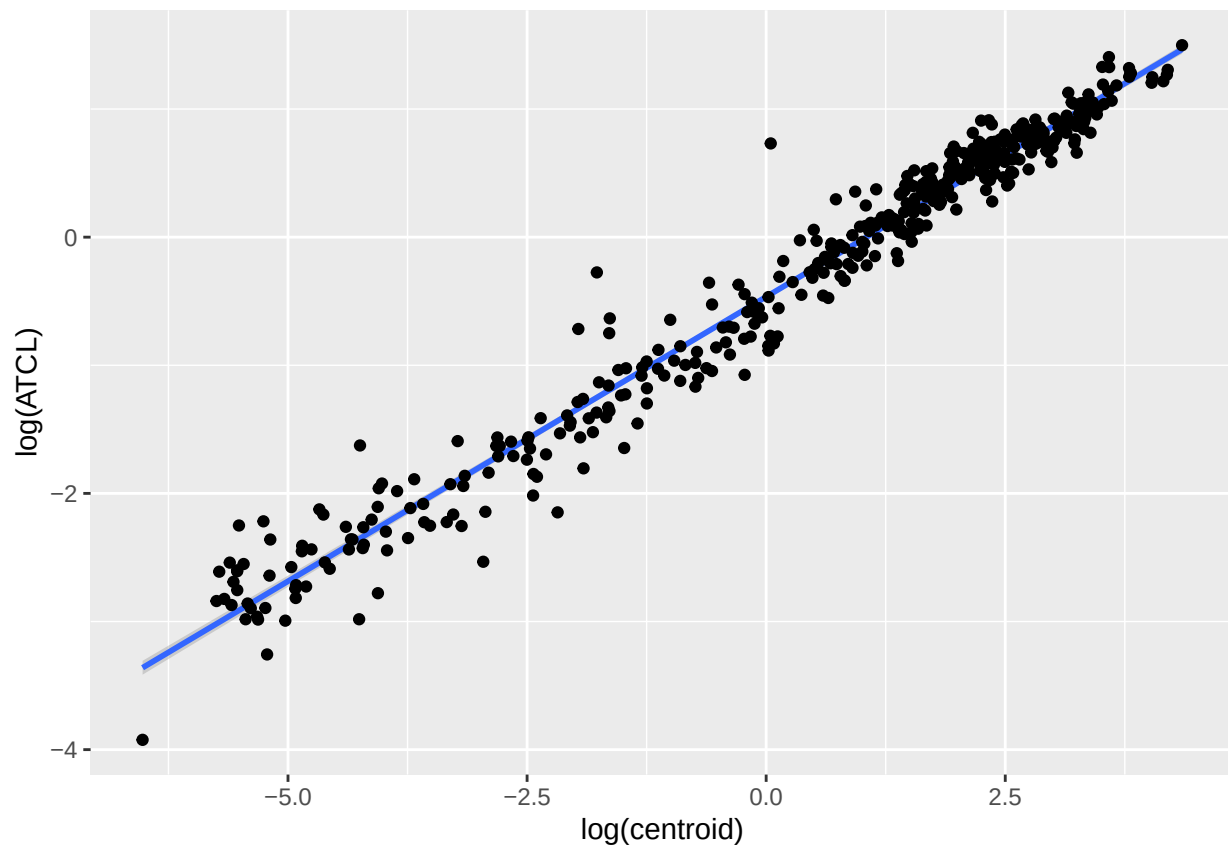
```
## Warning: Removed 18 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

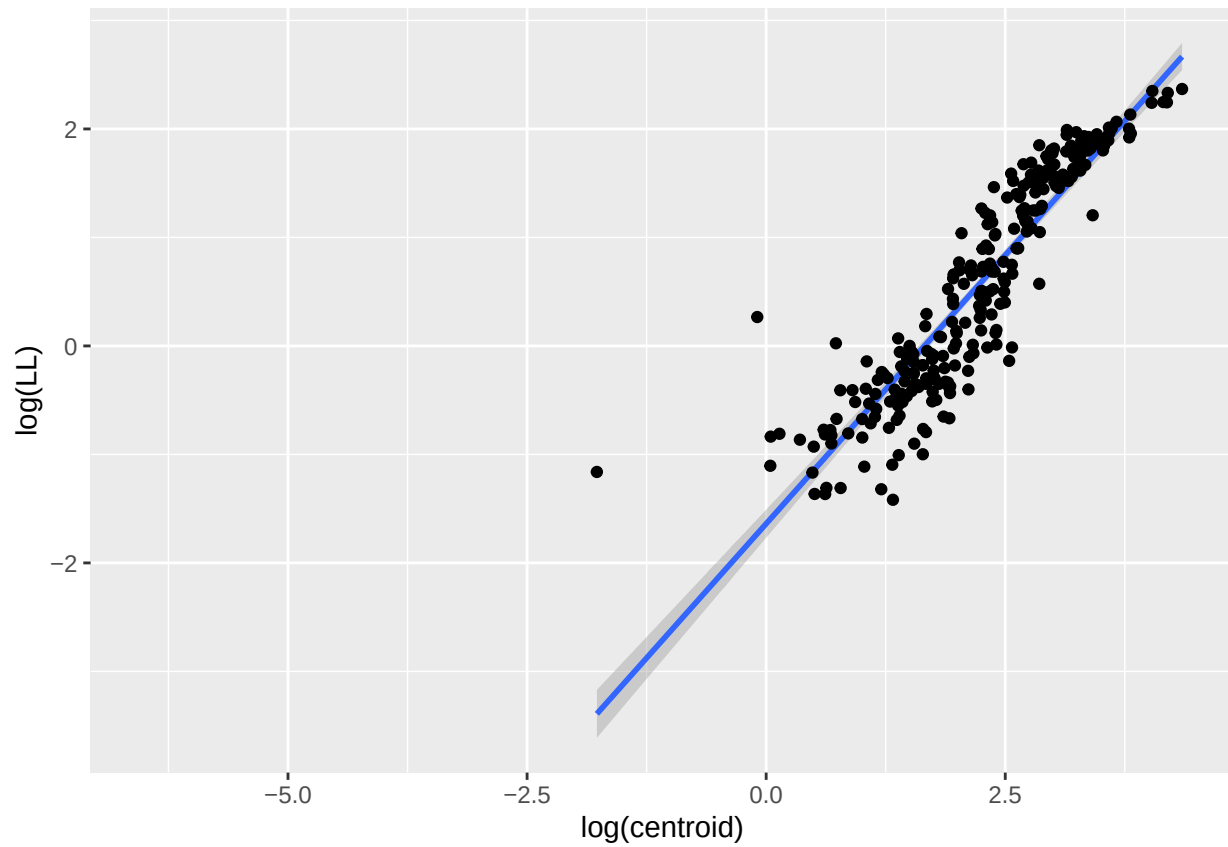
```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 181 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

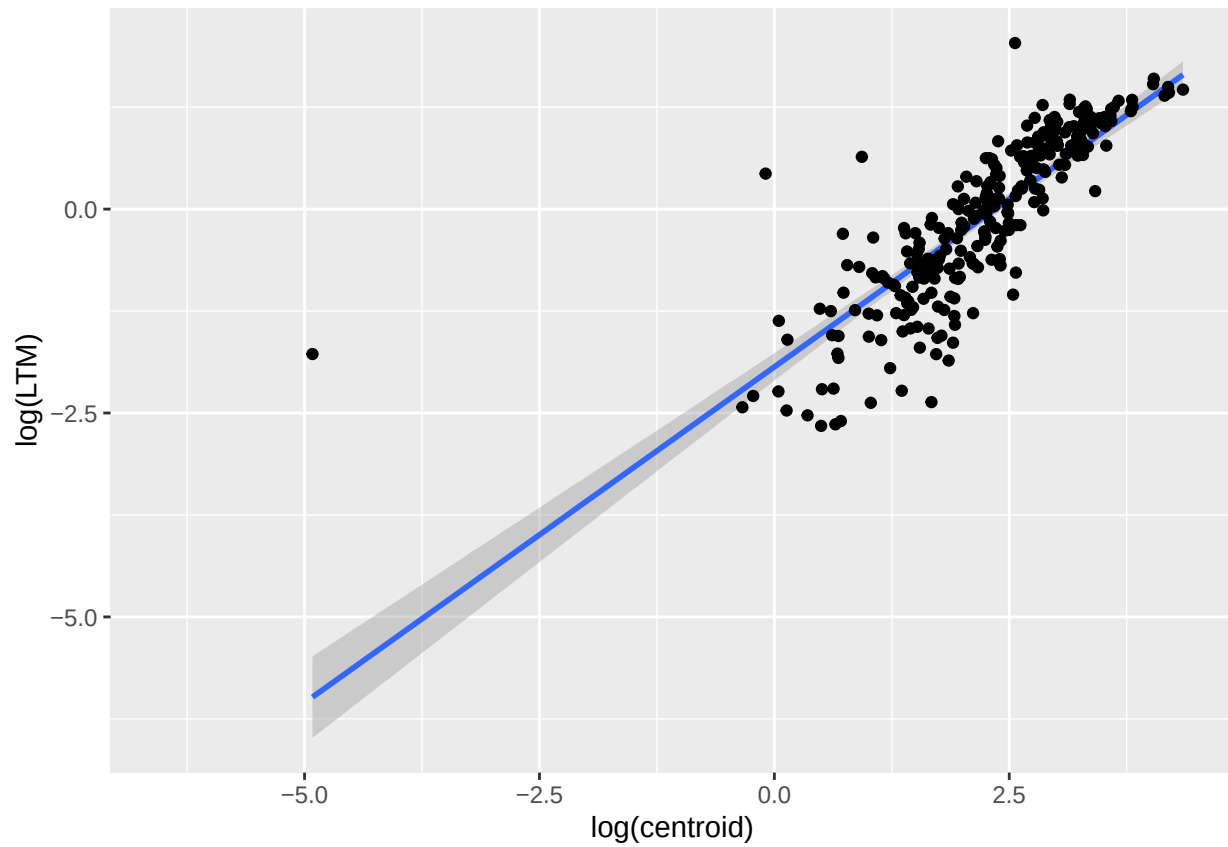
```
## Warning: Removed 181 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 183 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

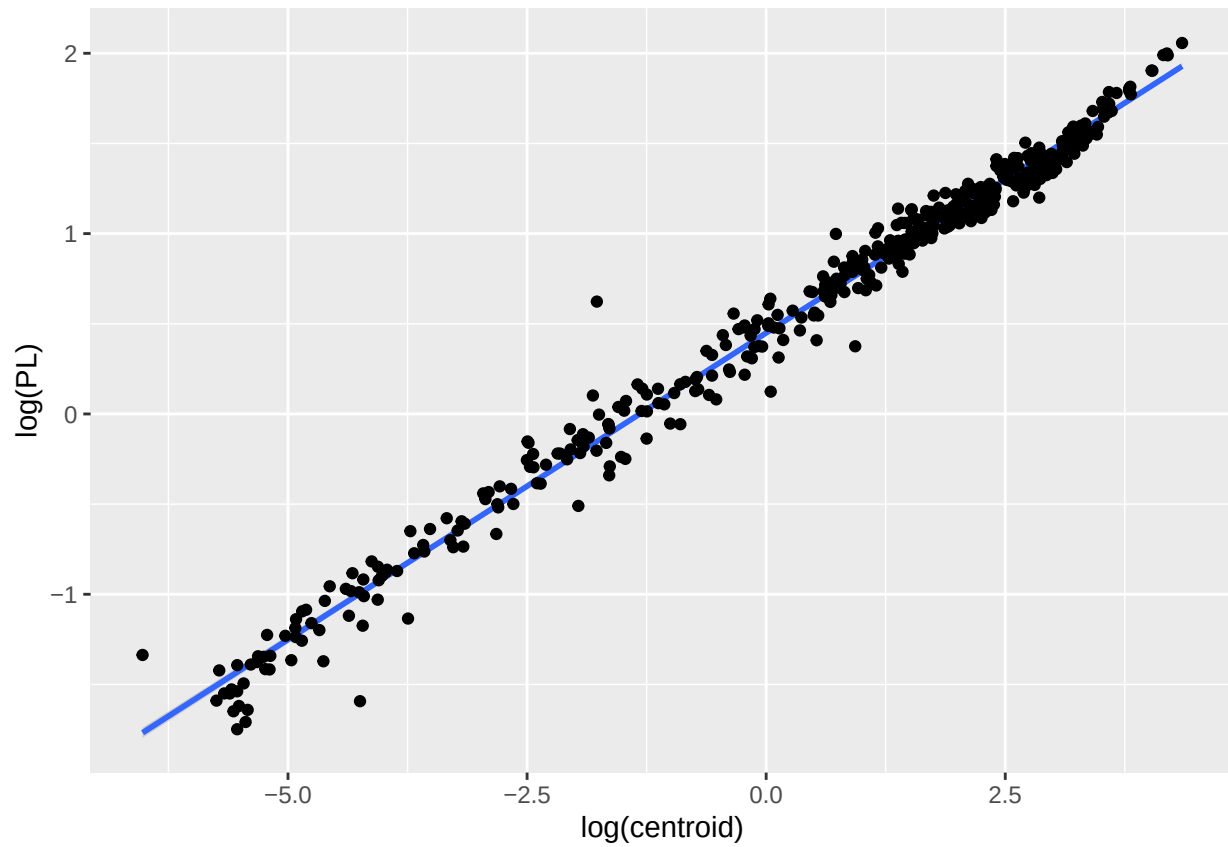
```
## Warning: Removed 183 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

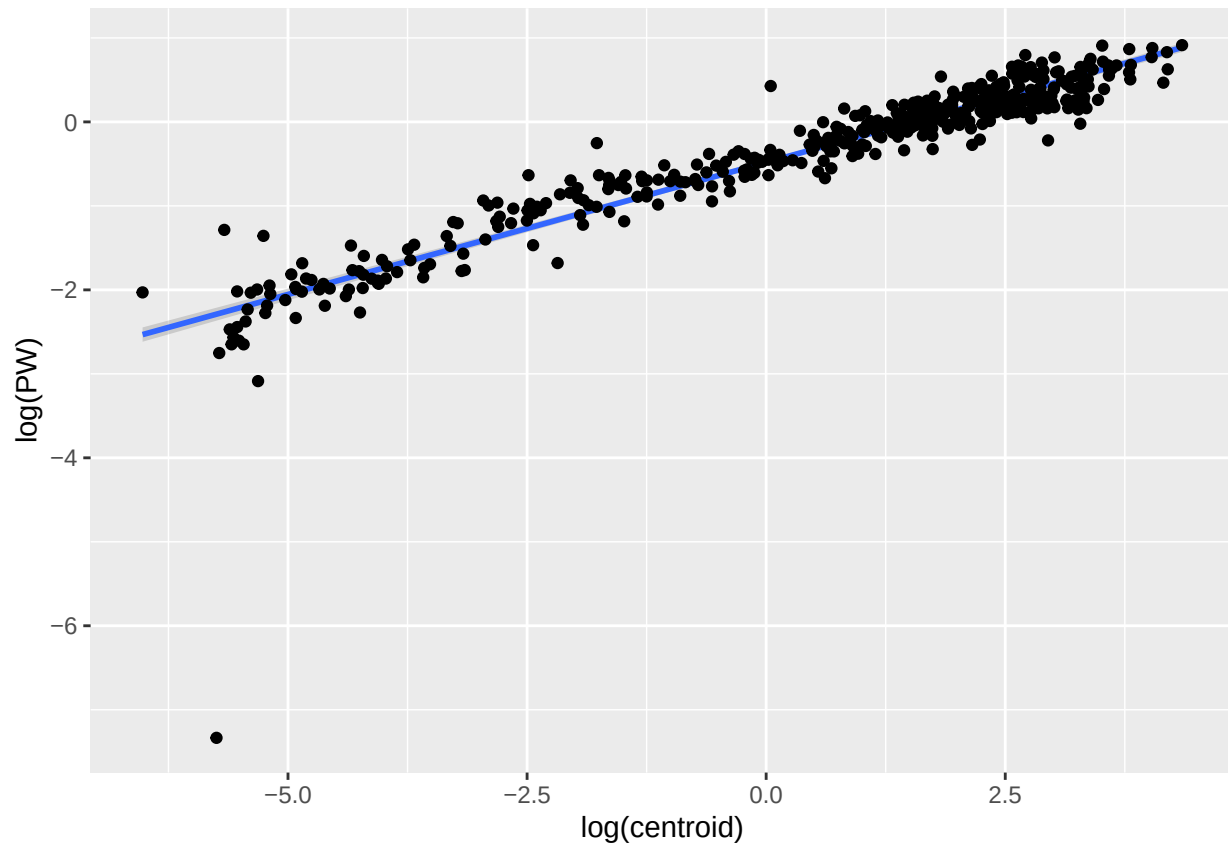
```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 18 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

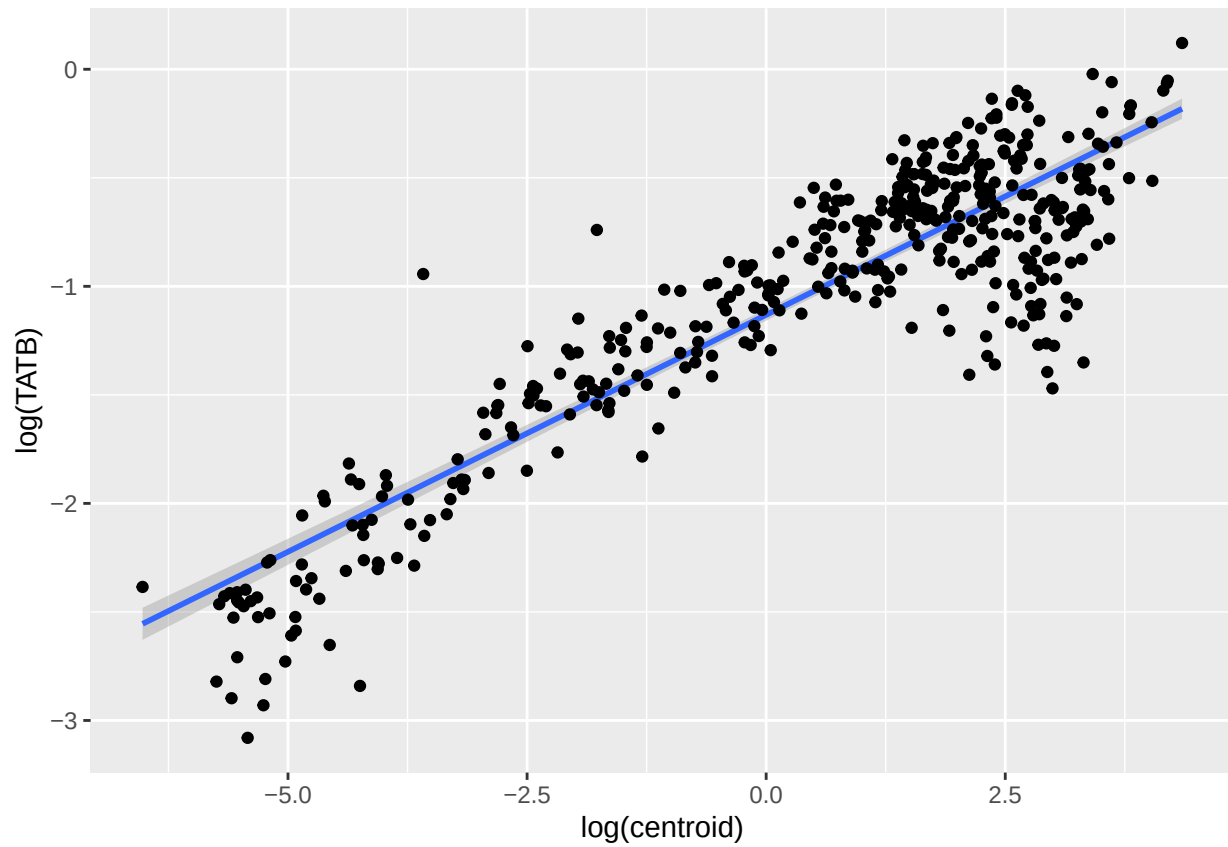
```
## Warning: Removed 18 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

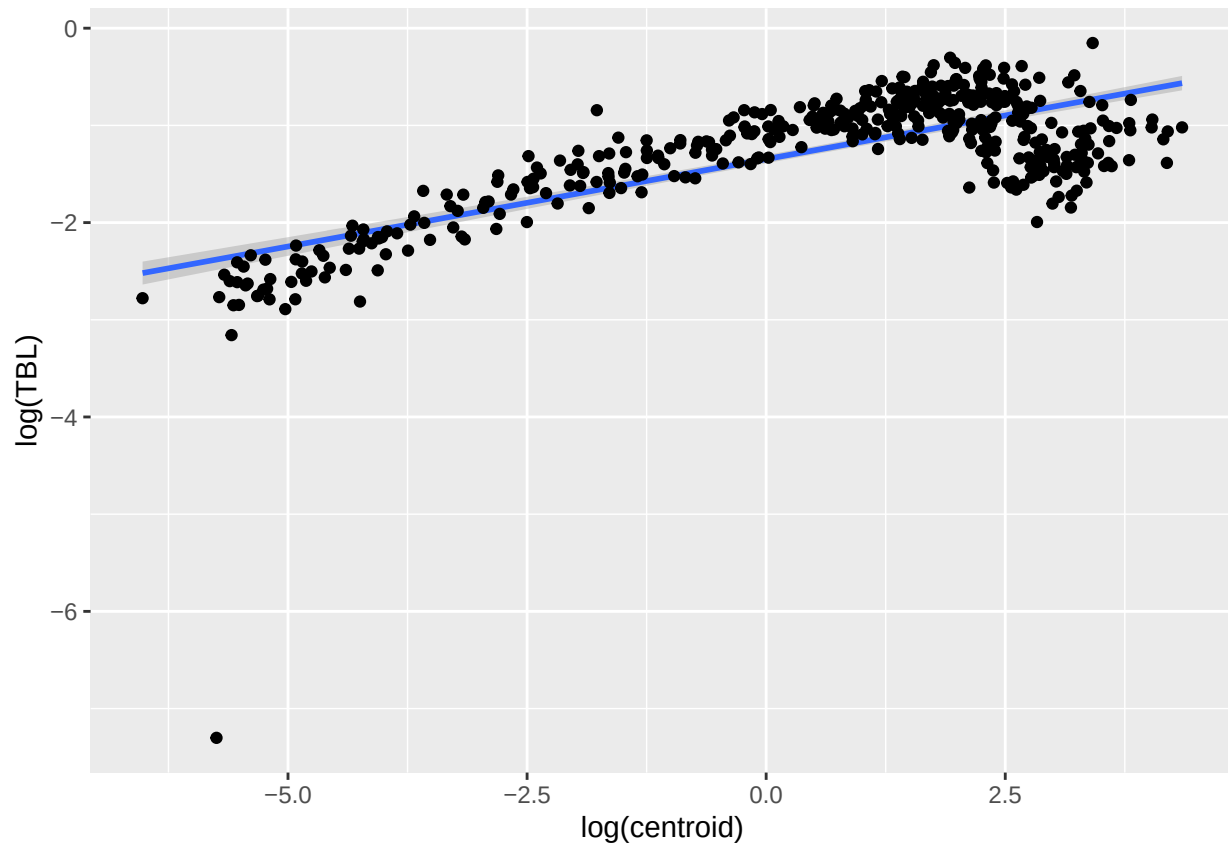
```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 18 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

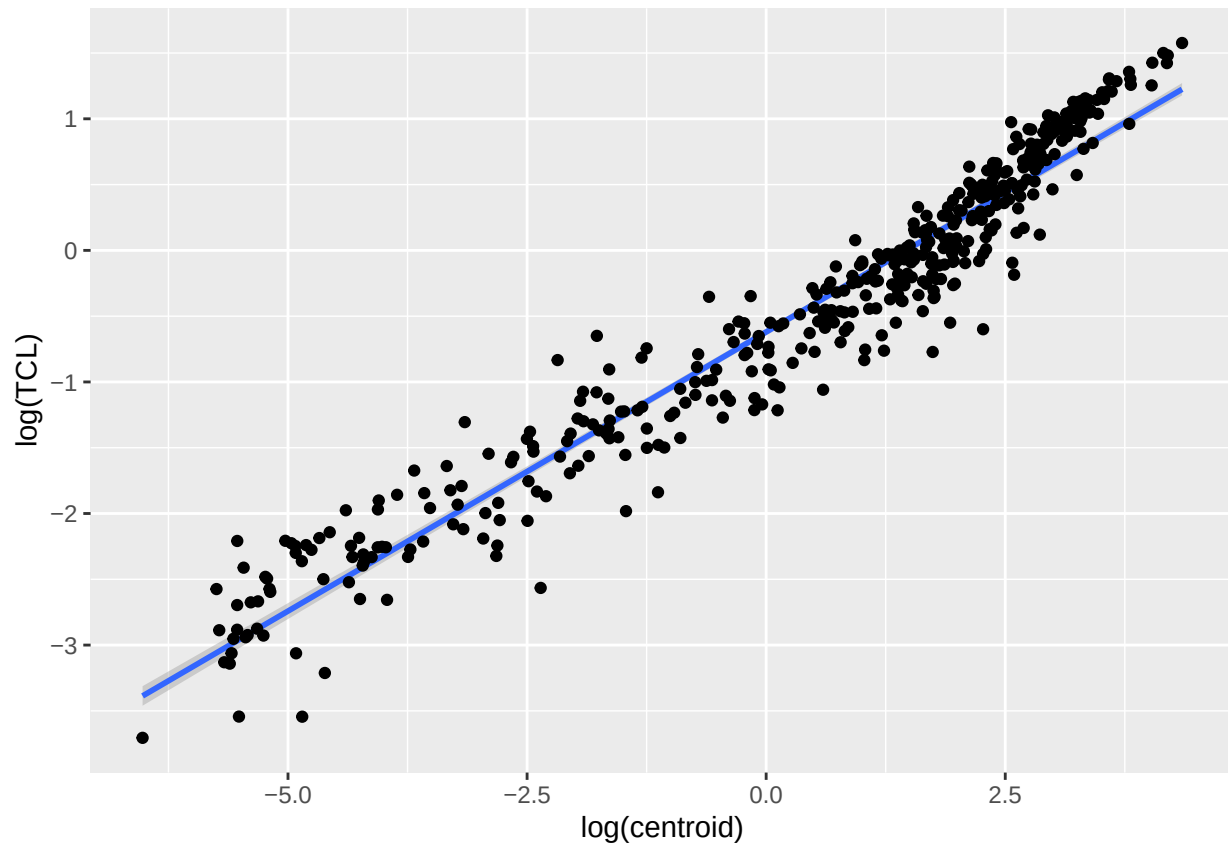
```
## Warning: Removed 18 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range ('stat_smooth()').  
## Removed 9 rows containing missing values or values outside the scale range  
## ('geom_point()').
```